Synthesis and Optimization of Isopropyl Xanthic Anhydride (iPrXA): Reactions of Potassium Isopropyl Xanthate with Ethyl Chloroformate

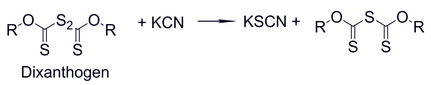
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Background

- iPrXA was first made by Welde in 1877 by reactions of potassium ethyl xanthate with ethyl chloroformate in water[1].
- iPrXA is an interesting compound for current and future uses in organosulfur chemistry and Professor Barany's research[2,3].
- iPrXA is used for mineral flotation[4], and as a softening and plasticizing agent[5].
- Other methods to make iPrXA include:
  - ❖ Reactions of an alkyl dixanthogen with potassium cyanide[6,7]:

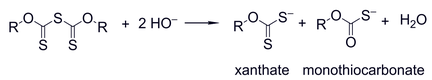


  - ❖ Reactions of potassium ethyl xanthate with phosgene in toluene[2,8].
  - ❖ Reactions of 2-chloro-4,5-dihydroimidazole with potassium alkyl xanthate[9]:

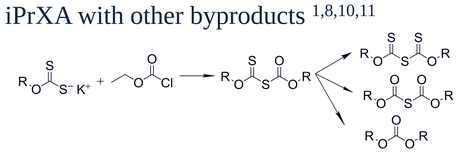


Statement of Problem

- iPrXA decomposes quickly into xanthate and monothiocarbonate in basic or alkaline solutions[6] (pH > 7)



- Reactions of potassium isopropyl xanthate with ethyl chloroformate will generate iPrXA with other byproducts [1,8,10,11]

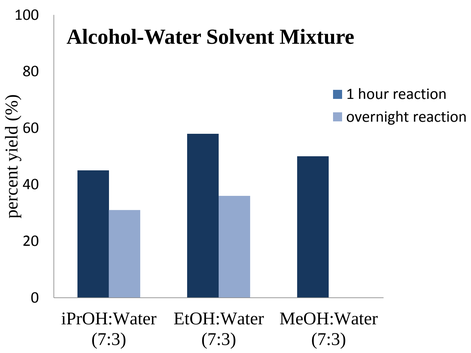


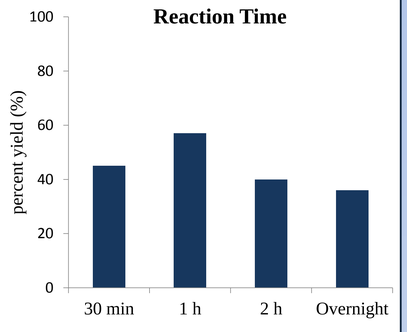
- In previous work, iPrXA was obtained at yields only up to 57%.

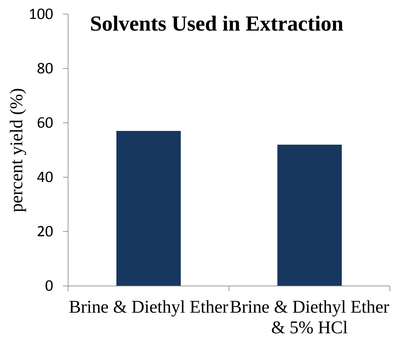
Significance of Study

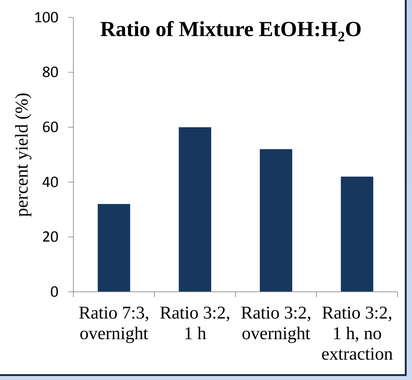
- Determine mechanism of reactions of potassium isopropyl xanthate and ethyl chloroformate to form isopropyl xanthic anhydride (iPrXA).
- Invent a new procedure to optimize and purify iPrXA.

Results



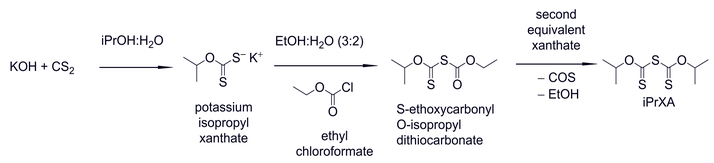






Experiments

- *Isopropyl Xanthic Anhydride (iPrXA):* Ethyl chloroformate (25 mL, 0.26 mol) was added over 15 min to a vigorously stirred solution of potassium *iso*-propyl xanthate (91.5 g, 0.53 mol) in EtOH–H_2O (3:2, 0.7 L) at 25 °C (negligible spontaneous exotherm). Stirring continued for an additional 1 h. Next, brine (0.5 L) and diethyl ether (0.5 L) were added, the organic layer was separated, dried ($MgSO_4$) and concentrated, and a pale yellow solid mass (58.4 g) including salts was obtained. NMR ($CDCl_3$): δ5.77 (septet), δ1.45 (doublet). The yellow solid was recrystallized with 300 mL hot hexane, and 37.5 g (61% yield) yellow crystal iPrXA was collected, melting point 52-54 °C.

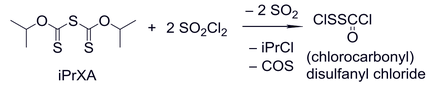


Conclusion

- Isopropyl xanthic anhydride, iPrXA, was the final product of the reactions of potassium isopropyl xanthate and ethyl chloroformate in an ethanol-water mixture. This reaction generated an S-ethoxycarbonyl O-isopropyl dithiocarbonate intermediate first, and then it reacted further with xanthate to give iPrXA[2,3].
- Many factors affected the yield of iPrXA: time of reaction, type of alcohol-water solvent mixture, and the ratio of alcohol and water in the mixture.
- The yield of iPrXA decreased as the time of reaction increased, and a time of 1 h gave an optimal yield.
- Ethanol-water mixture with the ratio 3:2 was chosen for this experiment since it gave the highest amount of iPrXA, and the iPrXA was pure after recrystallization.
- The best yield of this experiment was 61%.
- Some iPrXA was lost during extraction and recrystallization. The yield of iPrXA in the experiment could be increased if some loss during extraction and recrystallization could be eliminated.

Future Research

- iPrXA is a necessary reagent to make (chlorocarbonyl)disulfanyl chloride, an interesting compound in organosulfur chemistry[3]:



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